

Supersymmetric Topological Sigma Models and Doubling Spaces

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Preprint available at [arXiv:2507.08181](https://arxiv.org/abs/2507.08181)
(QR code links to abstract)

Supersymmetric Topological Sigma Models?

Witten's dictionary¹ between algebraic geometry and string theory:

Physics	Algebraic Geometry
Spacetime Geometry	projective $\mathbb{C}$ -variety
B-model spectrum	Dolbeault cohomology
A-model spectrum	De Rham cohomology
B-model D-branes	Derived category $D^b(X)$
A-model D-branes	Fukaya category $\text{Fuk}(X)$

The string theory “knows” about the derived and the Fukaya categories.

¹Witten arXiv:hep-th/9112056

Doubling Spaces?

Siegel's doubling space: an alternative method for computing string theories in torus geometry.²

Definition

Let X be a torus, $\hat{X}$ the dual torus (dual metric, dual complex/algebraic structure, etc.). Then, the doubling space for X is the Cartesian product

$$\mathbb{X} := X \times \hat{X}$$

Motivation: T-duality is a (large) diffeomorphism of $\mathbb{X}$.

Any string theory into X induces a “doubled theory” into $\mathbb{X}$.

²Siegel arXiv:hep-th/9305073 and hep-th/9302036, Hull and Zwiebach arXiv:0904.4664

Question

Can we extract algebraic geometry information about a manifold using doubling spaces?

Symplectic side: yes, holomorphic Lagrangian intersection cohomology in $\mathbb{X}$ encodes Fukaya category information³

Complex side: this talk

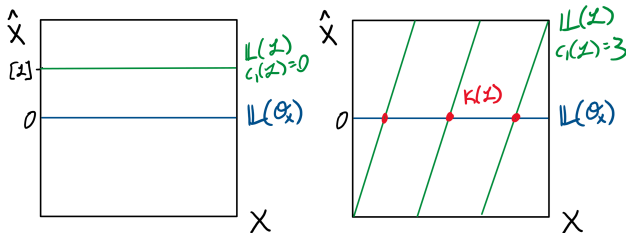
³Y. Qin arXiv:2207.10647

Framework

Definition

For any complex line bundle $\mathcal{L}$ on X there is a unique Lagrangian subtorus $\mathbb{L}(\mathcal{L})$ in $\mathbb{X}$ called the *lift* of $\mathcal{L}$.

$$\mathbb{L}(\mathcal{L}) = \{(x, \hat{x}) \in \mathbb{X} \mid \langle \hat{x}, \gamma_x \rangle = (-1)^{\xi(\gamma_x)} \text{hol}_{\nabla}(\gamma_x), \forall \gamma_x \in \pi_1(X, x)\}$$



Pictured: Elliptic curve case

Main Result I: Ext-groups

“Hom-spaces in $D^b(X)$ are computed in the doubling space”

Theorem (DMH, arXiv:2507.08181)

If $c_1(\mathcal{L}_1) = c_1(\mathcal{L}_2)$, then for all k

$$HF_{\mathcal{J}}^k(\mathbb{L}(\mathcal{L}_1), \mathbb{L}(\mathcal{L}_2)) \cong \text{Ext}^k(\mathcal{L}_1, \mathcal{L}_2)$$

where $HF_{\mathcal{J}}$ is Y. Qin’s “ $\mathcal{J}$ -holomorphic” Lagrangian Floer Cohomology.

Remark: The assumption on the Chern classes can be dropped, with a bit of work⁴.

⁴DMH, in progress

Main Result II: Generalized Complex Geometry

Definition (Hitchin, Gualtieri)

A *generalized complex structure* is an endomorphism $\mathcal{J}$ of $TX \oplus T^*X$ which squares to -1 (plus an integrability condition).

- Complex structures are GC structures: $\mathcal{J} = J \oplus J^*$
- Symplectic structures are GC structures: $\mathcal{J} = \omega \oplus \omega^{-1}$
- Can be a mix of these two, B-field transform, etc.

Main Result II: Generalized Complex Geometry

“Doubling spaces geometrize GC geometry”

Theorem (DMH, arXiv:2507.08181)

Any generalized complex structure $\mathcal{J}$ on X lifts to a unique complex structure on $\mathbb{X}$ compatible with the projection map.

Theorem (DMH, arXiv:2507.08181)

Furthermore, for a (sufficiently nice) generalized complex subtorus with $U(1)$ connection (M, ∇) , the lift $\mathbb{L}(M)$ is a $\mathcal{J}$ -holomorphic Lagrangian subtorus, and the generalized tangent space of (M, ∇) can be identified with the tangent space of $\mathbb{L}(M)$.

Conclusion

Results

- Line bundles lift to holomorphic Lagrangian subtori in $\mathbb{X}$
- Generalized complex structures lift to complex structures in $\mathbb{X}$
- Holomorphic Lagrangian intersection theory computes categorical hom-spaces (Ext or HF)

Applications

- coisotropic brane hom-spaces (Y. Qin)
- Hyperkähler (B, B, B) and (B, A, A) structures are “the same”
- Candidate for cohomology groups for generalized complex branes
- mirror symmetry?

Future Work

- Dropping the $c_1(\mathcal{L}_1) = c_1(\mathcal{L}_2)$ condition (expected)
- Lifting torsion sheaves and higher-rank vector bundles
- Doubling spaces for more general manifolds (K3 case is in progress, Mukai's moduli of stable sheaves?)
- Incorporating generalized BRST cohomology ⁵ to the doubled space
- Interaction with hyperkähler structures

⁵Kapustin and Li arXiv:hep-th/0501071

Thank you!

